

Aufgabe 1

a)

$$\vec{p_1 p_2} = (x_1, y_1, z_1)^T$$

$$\vec{p_1 p_3} = (x_2, y_2, z_2)^T$$

$$p_1 p_3 \times p_1 p_2 = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \times \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} y_2 z_1 - z_2 y_1 \\ z_2 x_1 - x_2 z_1 \\ x_2 y_1 - y_2 x_1 \end{pmatrix}$$

$$R_2 \times R_x = \frac{1}{|p_1 p_2|} \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \times \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \frac{1}{|p_1 p_2|} \begin{pmatrix} y_1 c - z_1 b \\ z_1 a - x_1 c \\ x_1 b - y_1 a \end{pmatrix}$$

$$R_2 = \frac{(x_1, y_1, z_1)^T}{\sqrt{x_1^2 + y_1^2 + z_1^2}} = \frac{(x_1, y_1, z_1)^T}{|p_1 p_2|} = \left(\frac{x_1}{|p_1 p_2|}, \frac{y_1}{|p_1 p_2|}, \frac{z_1}{|p_1 p_2|} \right)^T$$

$$R_x = \frac{p_1 p_3 \times p_1 p_2}{|p_1 p_3 \times p_1 p_2|} = \left(\frac{y_2 z_1 - z_2 y_1}{|p_1 p_3 \times p_1 p_2|}, \frac{z_2 x_1 - x_2 z_1}{|p_1 p_3 \times p_1 p_2|}, \frac{x_2 y_1 - y_2 x_1}{|p_1 p_3 \times p_1 p_2|} \right)^T = (a, b, c)^T$$

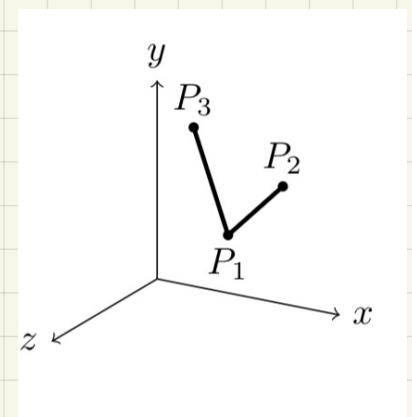
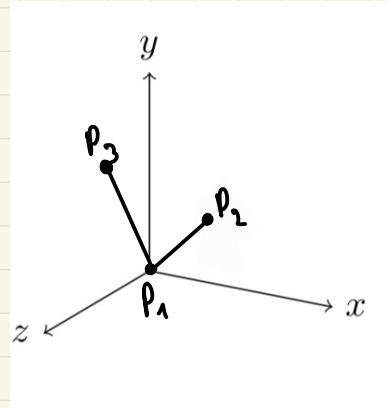
$$R_y = R_2 \times R_x = \left(\frac{y_1 c - z_1 b}{|p_1 p_2|}, \frac{z_1 a - x_1 c}{|p_1 p_2|}, \frac{x_1 b - y_1 a}{|p_1 p_2|} \right)^T$$

$$R = \begin{pmatrix} \frac{x_1}{|p_1 p_2|} & \frac{y_1}{|p_1 p_2|} & \frac{z_1}{|p_1 p_2|} \\ a & b & c \\ \frac{y_1 c - z_1 b}{|p_1 p_2|} & \frac{z_1 a - x_1 c}{|p_1 p_2|} & \frac{x_1 b - y_1 a}{|p_1 p_2|} \end{pmatrix}$$

$$b) \quad \vec{P_1 P_2} = (x_1, y_1, z_1)$$

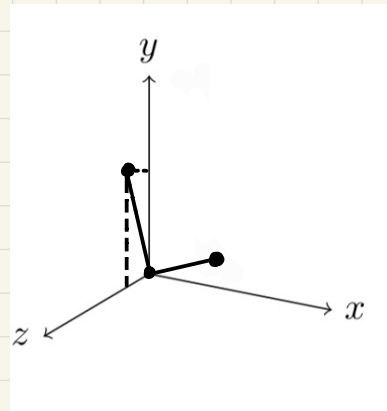
1: Verschiebung in den Ursprung

$$T(-x_1, -y_1, -z_1) = \begin{pmatrix} 1 & 0 & 0 & -x_1 \\ 0 & 1 & 0 & -y_1 \\ 0 & 0 & 1 & -z_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



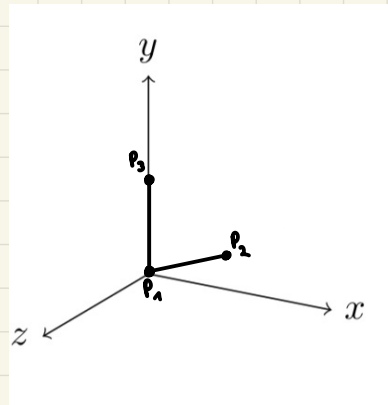
2: Rotation auf y-z-Ebene über z-Achse

$$R_2(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



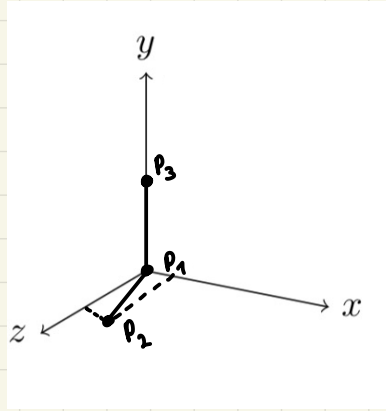
3: Rotation auf y-Achse über x-Achse

$$R_x(\theta') = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta' & -\sin \theta' & 0 \\ 0 & \sin \theta' & \cos \theta' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



4: Rotation auf x - z -Ebene über y -Achse

$$R_y(\theta'') = \begin{pmatrix} \cos \theta'' & 0 & -\sin \theta'' & 0 \\ 0 & 1 & 0 & 0 \\ \sin \theta'' & 0 & \cos \theta'' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



$$\Rightarrow M = R_y(\theta'') \cdot R_x(\theta') \cdot R_z(\theta) \cdot T$$

c) Rigid Body Transformation

Kombination aus:

- Translation t
- Rotation R

$$\Rightarrow M_{\text{rigid}} = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \quad \text{mit,}$$

$$R = R_y(\theta'') \cdot R_x(\theta') \cdot R_z(\theta) \quad (3 \times 3 \text{ Matrix ohne } w\text{-Komponente})$$

$$t = T's \text{ Translations teil}$$